

Razvan Nicolae

571-353-5612 | razvannicolae@outlook.com | [linkedin.com/in/razvannicolae](https://www.linkedin.com/in/razvannicolae) | github.com/razvannicolae

EDUCATION

University of Virginia

Bachelor of Science, Computer Science and Mathematics

Charlottesville, VA

May 2029

- Relevant Coursework: Data Structures and Algorithms, Software Development, Computer Systems, Discrete Math, Probability, Linear Algebra, Differential Equations, Calculus III
- Activities: Alternative Investment Fund, ML @ UVA, HooHacks, Google Developer Student Club, Science Olympiad
- Cumulative GPA: 4.0/4.0 | SAT: 1560/1600

EXPERIENCE

Software Developer Intern

Booz Allen Hamilton

Jun. 2026 – Present

McLean, VA

- Built a reusable real-time safety gate for LLM-driven humanoids, validating the **task-planning agent's** intents and **execution-layer** motion commands at **2** checkpoints with deterministic rules and **AWS Bedrock** judging
- Shipped a **React/TypeScript** operator console (**10** panels) to author tighten-only **policies** and run replays, backed by **OpenTelemetry/Phoenix, Postgres, and MinIO**, with drop-in config for any **AWS Strands** agent
- Built the **execution gateway** routing **9** semantic actions to **A*** path-planning or **RL** control, hot-swapping **3** policy backends including **CPU TorchScript/ONNX** inference, with pre-dispatch validation and a **5-service Docker** deploy

Computer Vision Research Assistant

University of Virginia Engineering Link Lab

Jan. 2026 – Jun. 2026

Charlottesville, VA

- Built **AriaData**, a unified Python interface over Meta **Project Aria's** SDK and **Machine Perception Services**, loading eye-gaze, hand-tracking, and SLAM streams into timestamp-indexed **pandas** DataFrames for frame-aligned access
- Added lossless **JSON serialization** and a **Rerun** 3D timeline viewer for gaze points, 21-landmark hand poses, and wrist/palm trajectories, giving the lab fast recording replay and validation
- Integrated hand-tracking data with **Robosuite**, mapping a human wrist's 3D pose onto a simulated **Franka Panda** arm and rendering offscreen MP4 runs to test real-to-sim manipulation

Research Intern

Howard Hughes Medical Institute | Janelia Research Campus

Jun. 2025 – Aug. 2025

Ashburn, VA

- Engineered a **reproducible 3D object-tracking system** for neuroscience experiments using a UR3E robotic arm and four Emergent Vision cameras across cross-functional research teams
- Developed an **OpenCV camera-calibration pipeline** with ChArUco boards and sparse-bundle adjustment, achieving **sub-0.1 pixel re-projection error** for accurate 3D triangulation
- Trained and deployed a **YOLOv8n** model on a 1,000+ image dataset at **sub-2ms inference** on GPU, automating the end-to-end training pipeline in Python to cut workflow time by **20%**

Software Engineering Intern

Fuel and Tire Saver Systems Company LLC.

Jun. 2024 – Jun. 2025

Sterling, VA

- Redesigned an internal **JavaScript web application** with a custom algorithm that eliminates message-logging redundancy, improving performance and usability while cutting **memory usage by 10%**
- Sustained **95%+ uptime** on mission-critical systems by diagnosing and resolving SQL synchronization, VirtualBox KVM process errors, and payment-processing server failures
- Built a **socket-based communication system** streaming real-time battery-voltage and nitrogen-concentration metrics from 5+ field devices to an internal administrative dashboard

PROJECTS

Baby Diagnostics | [GitHub](#) | *React, TypeScript, FastAPI, OpenCV, PostgreSQL*

Apr. 2026

- Built a **React + FastAPI** mobile web app that lets NICU staff scan diagnostic test strips by phone, using an **OpenCV** pipeline to match median **HSV** values against calibrated references for pH, creatinine, and vitamin readings
- Secured patient data with **JWT** auth, **AES-256** PII encryption, and in-memory image handling that persists only SHA-256 hashes, streaming **Ollama** result explanations over **WebSocket**

Food Facts MCP Server | [GitHub](#) | *Python, FastMCP, SQLite, REST APIs*

Mar. 2026

- Built a **Model Context Protocol** server exposing **13 tools** that serve AI assistants real-time, citation-backed nutrition data from the **USDA FoodData Central** and **FatSecret REST APIs** across 4M+ foods
- Implemented an **OAuth2** API client and a **SQLite** caching layer that cut repeat-query latency from 1–2s to **sub-1ms**, hardened by a **69-test** suite spanning tools, cache, and HTTP transport

Digital Human | [GitHub](#) | *React, TypeScript, Node/Express, WebSocket, Groq*

Feb. 2026

- Built a real-time conversational **avatar** system that lets users talk to an AI digital human, orchestrating **Groq** (Llama 3.3-70B), **ElevenLabs** speech synthesis, and **Tavus**/Daily.co video into one streaming pipeline
- Engineered a **WebSocket-driven Node.js/Express** backend streaming each turn end-to-end (LLM to speech to lip-sync) for low-latency replies, with a **React + TypeScript** frontend managing session state and live chat

Consumer Sentiment Analysis | [GitHub](#) | *Python, Apify, HuggingFace, Numpy, Pandas*

Nov. 2025

- Built a **multi-platform data pipeline** with the Apify API to scrape and aggregate **6,000+ reviews** from Amazon, Instagram, and TikTok across four competitor brands, automating cleaning and brand classification
- Deployed a **CardiffNLP RoBERTa** transformer through HuggingFace for sentiment classification, generating automated visualizations and brand-insight reports with platform-specific engagement metrics

Muscle Metrics | [GitHub](#) | *Python, Numpy, OpenPose, Arduino*

Mar. 2025

- Engineered a **biomechanics analysis system** with OpenPose and **sEMG sensors**, training CNNs to predict myoelectric patterns across five muscles and classify exercise form at **75% accuracy**
- Processed and labeled **multimodal sensor data** with NumPy and Arduino microcontrollers, building a **10,000+ frame** training dataset of paired motion-capture and biosignal recordings

ACTIVITIES

Analyst <i>Alternative Investment Fund</i> - Selected for a competitive student-run fund (4% acceptance) managing \$60K AUM across fundamental equities, special situations, global macro, and quantitative strategies through a team research process - Authored a long investment thesis on SharkNinja (pitch deck, <u>financial model</u>) backed by bottoms-up modeling and competitive sentiment analysis, with shares up over 50% since the pitch	Aug. 2025 - Present Charlottesville, VA
---	--

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL, HTML/CSS, MATLAB
Frameworks & Libraries: React, Node.js/Express, FastAPI, Django, PyTorch, TensorFlow, ONNX, OpenCV, pandas, NumPy
Tools & Platforms: AWS, Docker, PostgreSQL, SQLite, Git/GitHub, Linux, REST/WebSocket, CI/CD, Agile, Agentic AI/LLMs